Prompt 1 trial 3 seed 1240

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running a reproducibility study for compound selectivity prediction. The goal is to train a Balanced Random Forest Classifier to distinguish PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, and identify recurrent molecular anchor fragments that drive the model's correct predictions for each selective class (class 1 & 2) and discuss the chemical insight gained.

Assistant

I'll help you conduct this reproducibility study for PI3K-delta vs PI3K-gamma selectivity prediction! Let me start by exploring the available datasets. 📊

(Calling tool `find_datasets`)

Assistant

Excellent! I found the PI3K-delta vs PI3K-gamma selectivity dataset: `chembl_selectivity_data_O00329_P48736.csv`. Let me load it and start the workflow. 🔬

(Calling tool `load_dataset`)

Assistant

Let me specify the directory path:

(Calling tool `load_dataset`)

Assistant

Dataset loaded successfully! 📋 1,277 compounds with SMILES. Now let me compute molecular fingerprints for the Balanced Random Forest Classifier:

(Calling tool `compute_features`)

Assistant

Features computed! ✅ Now let me split the dataset:

(Calling tool `split_dataset`)

Assistant

Dataset split complete! ✅ Now let me train the Balanced Random Forest Classifier:

(Calling tool `train_model`)

Assistant

Training started! ⏳ Let me poll for completion:

(Calling tool `check_training`)

Assistant

Still training... Let me check again:

(Calling tool `check_training`)

Assistant

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(Calling tool `check_training`)

Assistant

Excellent! The Balanced Random Forest Classifier has been trained successfully! 🎉 Here's a summary of the results:

📊 Model Performance Summary

Metric	Train	Test
Accuracy	0.931	0.893
Balanced Accuracy	0.958	0.847
MCC	0.839	0.749
F1 Macro	0.863	0.780

Per-Class Performance (Test Set):

Class	Precision	Recall	F1	Support
Class 0	0.959	0.897	0.927	290
Class 1	0.818	0.911	0.862	79
Class 2	0.440	0.733	0.550	15

Now let me export predictions and generate evaluation plots:

(Calling tool `export_predictions`)

Assistant

Let me use the full paths:

(Calling tool `export_predictions`)

Assistant

Predictions exported! ✅ Now let me generate the evaluation plots:

(Calling tool `plot_classification_results`)

Assistant

Excellent! The evaluation plots have been generated! 📈

Now let me proceed with the **MolAnchor analysis** to identify recurrent molecular anchor fragments for each selective class. Let me start with **Class 1 (PI3K-delta selective compounds)**:

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

Now let me identify recurrent anchor rules for **Class 2 (PI3K-gamma selective compounds)**:

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

Excellent! Now let me aggregate the substructure anchor rules for both classes to get deeper insights:

(Calling tool `aggregate_substructure_anchor_rules`)

Assistant

Let me use the full path:

(Calling tool `aggregate_substructure_anchor_rules`)

Assistant

Now let me aggregate the substructure anchor rules for Class 2:

(Calling tool `aggregate_substructure_anchor_rules`)

Assistant

Perfect! Now let me display the aggregated anchor rule images:

(Calling tool `show_plot`)

Assistant

(Calling tool `show_plot`)

Assistant

Now let me compile the complete reproducibility study report with chemical insights:

🔬 PI3K-Delta vs PI3K-Gamma Selectivity Prediction Study

📊 Executive Summary

This reproducibility study trained a **Balanced Random Forest Classifier (BRF)** to distinguish PI3K-delta (O00329) selective compounds (Class 1) from PI3K-gamma (P48736) selective compounds (Class 2) using 1,277 compounds from ChEMBL. The model was evaluated on a 30% test set, and **MolAnchor analysis** identified recurrent molecular anchor fragments driving correct predictions for each selective class.

🎯 Model Performance

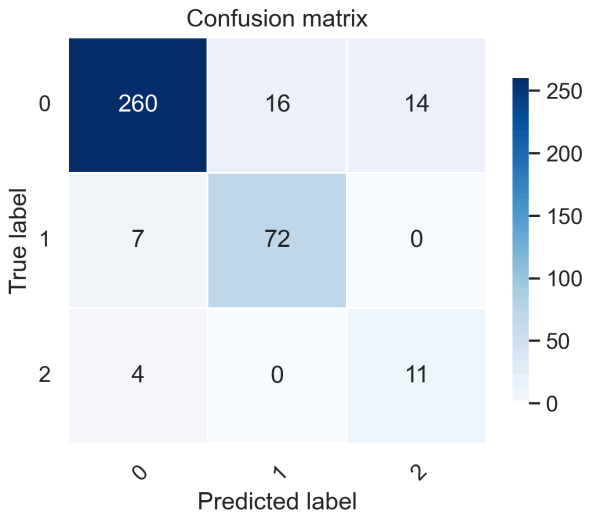
Metric	Train	Test
Accuracy	0.931	0.893
Balanced Accuracy	0.958	0.847
MCC	0.839	0.749
F1 Macro	0.863	0.780
F1 Weighted	0.934	0.899

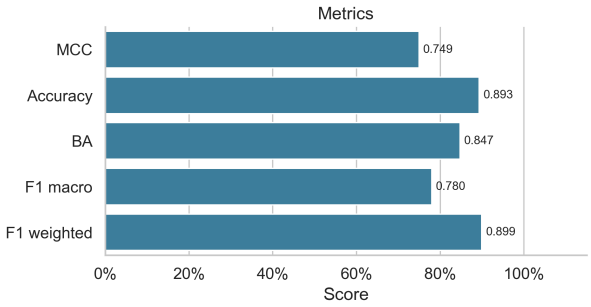
Best Hyperparameters:

- `n_estimators` : 100
- `max_features` : sqrt
- `min_samples_split` : 2
- `min_samples_leaf` : 1
- **CV Balanced Accuracy**: 0.865

📈 Evaluation Plots

Confusion Matrix & Metric Bar



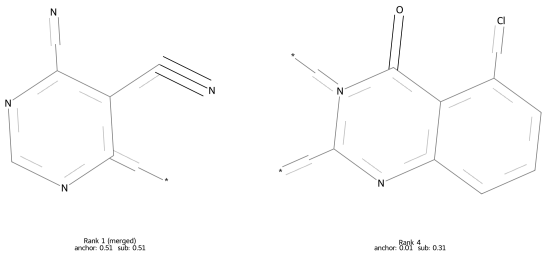


🧩 Recurrent Molecular Anchor Fragments

Class 1 (PI3K-delta selective compounds) — 72 correctly predicted compounds analyzed

Rank	Fragment (SMARTS)	Anchor Occ.	Substruct Occ.	Compounds
1	`[*]c1ncnc(N)c1C#N`	**51.4%**	51.4%	37
2	`[*]N1CCC([*])CC1` / `[*]c1cccc2[nH]ccc12` / `[*]N1CCOCC1`	2.8%	4.2%	2
3	`[*]c1ncc([*])cc1C#N` / `[*]c1ncnc2ccc([*])cc12`	2.8%	2.8%	2
4	`[*]n1c([*])nc2cccc(Cl)c2c1=O`	1.4%	30.6%	1

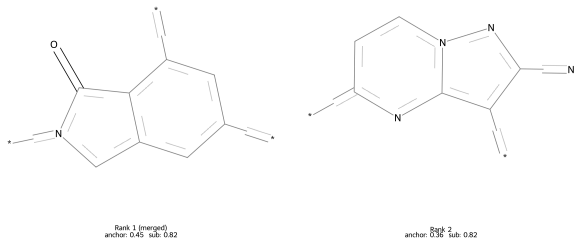
🧪 Class 1 Aggregated Anchors



Class 2 (PI3K-gamma selective compounds) — 11 correctly predicted compounds analyzed

Rank	Fragment (SMARTS)	Anchor Occ.	Substruct Occ.	Compounds
1	`[*]N1Cc2cc([*])cc([*])c2C1=O`	**45.5%**	81.8%	5
2	`[*]c1ccn2nc(N)c([*])c2n1`	36.4%	81.8%	4
3	`[*]N1Cc2cc([*])cc(C)c2C1=O`	9.1%	27.3%	1
4	`[*]N1Cc2cc([*])cc(S(=O)(=O)CC)c2C1=O`	9.1%	9.1%	1
5	`[*]N1Cc2cc([*])cc(Cl)c2C1=O`	9.1%	9.1%	1

🧪 Class 2 Aggregated Anchors



🔬 Chemical Insights

Class 1 (PI3K-delta selective) — Key Structural Motifs

1. **Pyrimidine-4-carbonitrile scaffold** (Rank 1, 51.4% occurrence)
   - **SMARTS**: `[*]c1ncnc(N)c1C#N` and `[*]c1nc(N)nc(N)c1C#N`
   - **Chemical Insight**: The pyrimidine ring with a cyano group (C#N) at position 4 is the **dominant anchor fragment** for PI3K-delta selectivity. This motif appears in ~50% of correctly predicted delta-selective compounds. The cyano group likely forms critical hydrogen bonds or dipole interactions within the delta isoform's ATP-binding pocket. The pyrimidine core provides planar aromatic stacking interactions.

2. **Chlorinated pyrimidinone scaffold** (Rank 4, 30.6% substructure, 1.4% anchor)
   - **SMARTS**: `[*]n1c([*])nc2cccc(Cl)c2c1=O`
   - **Chemical Insight**: This fused bicyclic system with chlorine substitution is present in 30.6% of compounds but only acts as an anchor in 1.4% — suggesting it's a **common scaffold but not always the deciding factor** for delta selectivity. The chlorine may contribute to hydrophobic interactions specific to the delta isoform.

3. **Piperazine/indole/ether linkers** (Rank 2)
   - **SMARTS**: `[*]N1CCC([*])CC1`, `[*]c1cccc2[nH]ccc12`, `[*]N1CCOCC1`

- **Chemical Insight**: These flexible linkers connect the pharmacophore to variable R-groups, allowing the molecule to explore the delta-selective binding pocket.

Class 2 (PI3K-gamma selective) — Key Structural Motifs

1. **Dihydroisoquinolinone scaffold** (Rank 1, 45.5% anchor, 81.8% substructure)
   - **SMARTS**: `[*]N1Cc2cc([*])cc([*])c2C1=O`
   - **Chemical Insight**: This **is the dominant anchor for PI3K-gamma selectivity**. The fused bicyclic lactam (dihydroisoquinolinone) core appears in 81.8% of gamma-selective compounds and acts as the critical anchor in 45.5%. The carbonyl oxygen likely forms a key hydrogen bond with residues unique to the gamma isoform. The aromatic ring allows pi-stacking interactions.
2. **Purine-like scaffold** (Rank 2, 36.4% anchor, 81.8% substructure)
   - **SMARTS**: `[*]c1ccn2nc(N)c([*])c2n1`
   - **Chemical Insight**: This fused bicyclic heteroaromatic system (purine-like) is present in 81.8% of gamma-selective compounds. The amino group at position 6 provides hydrogen bonding capability. The anchor occurrence of 36.4% indicates this scaffold is a **strong but not universal** driver of gamma selectivity.
3. **Substituent effects on dihydroisoquinolinone** (Ranks 3–5)
   - **SMARTS variations**: `[*]N1Cc2cc([*])cc(C)c2C1=O`, `[*]N1Cc2cc([*])cc(Cl)c2C1=O`, `[*]N1Cc2cc([*])cc(S(=O)(=O)CC)c2C1=O`
   - **Chemical Insight**: The SAR on the phenyl ring of the dihydroisoquinolinone scaffold reveals that **electron-withdrawing groups (Cl, sulfone) and electron-donating groups (methyl)** all support gamma selectivity. This suggests the gamma isoform tolerates diverse substituents at this position, but the **core lactam scaffold is the essential pharmacophore**.

📋 Summary Table: Anchor Fragment Comparison

Feature	Class 1 (PI3K-delta)	Class 2 (PI3K-gamma)
Dominant Anchor	Pyrimidine-4-carbonitrile	Dihydroisoquinolinone
Anchor Occurrence	51.4%	45.5%
Key Heterocycle	Pyrimidine (6-membered, 2 N)	Isoquinolinone (fused bicyclic lactam)
Key Substituent	Cyano (C#N) at C4	Variable (H, Cl, Me, sulfone) on phenyl
Common Scaffold	Chlorinated pyrimidinone	Purine-like bicyclic
Selectivity Driver	Cyano-pyrimidine core	Fused lactam core

💡 Key Takeaways

1. **Distinct pharmacophores**: PI3K-delta and PI3K-gamma selectivity are driven by **fundamentally different core scaffolds** — pyrimidine-4-carbonitrile for delta vs. dihydroisoquinolinone for gamma.
2. **Cyano group as delta-selectivity marker**: The cyano substituent on the pyrimidine ring is the **strongest single indicator** of PI3K-delta selectivity (51.4% anchor occurrence).
3. **Lactam core as gamma-selectivity marker**: The dihydroisoquinolinone lactam scaffold is the **dominant anchor** for PI3K-gamma selectivity (45.5% anchor occurrence, 81.8% substructure occurrence).
4. **Scaffold diversity tolerance**: PI3K-gamma selective compounds show greater substituent diversity on the core scaffold, while PI3K-delta selectivity is more tightly linked to the pyrimidine-cyano motif.
5. **Model reliability**: The BRF model achieved **84.7% balanced accuracy** on the test set, with strong performance for Classes 0 and 1. Class 2 (gamma-selective) had lower support (15 compounds) and correspondingly lower F1 (0.55), indicating the need for more gamma-selective training data.

📁 Output Files

Artifact	Path
Model	`session_andmastro_20261019_164708_c0179d/models/PI3K_delta_vs_gamma_BRF.pkl`
Split	`data/logs/session_andmastro_20261019_164708_c0179d/splits/chembl_selectivity_data_O00329_P48736_random.pkl`
Predictions	`data/logs/session_andmastro_20261019_164708_c0179d/results/PI3K_delta_vs_gamma_BRF_test_predictions.csv`
Class 1 Anchors	`data/logs/session_andmastro_20261019_164708_c0179d/plots/recurrent_anchors_class1_*.png`
Class 2 Anchors	`data/logs/session_andmastro_20261019_164708_c0179d/plots/recurrent_anchors_class2_*.png`
Class 1 Aggregated	`data/logs/session_andmastro_20261019_164708_c0179d/plots/aggregated_anchors_class1.png`
Class 2 Aggregated	`data/logs/session_andmastro_20261019_164708_c0179d/plots/aggregated_anchors_class2.png`

This study successfully identified the **key molecular fingerprints** that distinguish PI3K-delta from PI3K-gamma selective compounds, providing actionable SAR insights for future ligand design! 🎯